Coding Assignment 2

What can a traveller learn from a year of flights?

A travel editor wants to know how often flights from New York arrived late in 2013. Give the editor a table they can check. Then pitch a question of your own and answer it with one figure the editor could publish.

Individual take-home assignment 100 marks · 10% of the course grade · about three hours. Use pandas and plotnine from the Day 2 environment.

Open your pull request by **Monday, 16 November 2026 at 23:59 Vilnius time**.

Get the data

Download both files into a folder called `data` beside your notebook. Keep the original files so that someone can run your analysis again.

- **Download flights.csv.zip** for 336,776 scheduled flight records from Newark (EWR), Kennedy (JFK) and LaGuardia (LGA) in 2013. Keep this file zipped.
- **Download airlines.csv** to get the full airline `name` for each `carrier` code. If the CSV opens as text, save it as `airlines.csv`.

```
import pandas as pd
flights = pd.read_csv("data/flights.csv.zip")
airlines = pd.read_csv("data/airlines.csv")
```

Departure and arrival delays are measured in minutes against the schedule. Negative values mean early and zero means on time. The `hour` column is the scheduled departure hour. The data come from the U.S. Bureau of Transportation Statistics via Hadley Wickham's *nycflights13* (CC0). Michael Chow provides the CSV files. See the flight variable guide and airline lookup guide.

1. Build the analysis sample [20 marks]

(a) State what one row represents in each file. Show both shapes, check the type of `arr_delay`, and count exact duplicate flight rows across all original columns. Explain whether any rows need removing. An airline flight number can recur on different dates. [8]

(b) Show how many flight records have a known arrival delay and how many have a missing one. Check that the two counts add up to your flight total. Create a late-arrival indicator for delays of **15 minutes or more**. Keep the outcome missing when the delay is missing. Show how your rule handles an early arrival, 14 minutes, exactly 15 minutes and a missing value. [12]

2. Attach the airline names [10 marks]

Join the lookup to the flight records using `carrier`. Check that each lookup code occurs once, validate the many-to-one join in pandas, and show the row counts before and after. Report how many flights have an unmatched code. Use the airline names in your table on the next page.

3. Give the editor a table they can check [30 marks]

(a) Identify the five busiest airlines by their number of scheduled flight records. Present one row per airline, busiest first, then a final row for **all airlines combined**, including those outside the top five. Show the airline name, scheduled flights, flights with a known arrival delay, late arrivals, and the percentage late among flights with a known arrival delay. Give columns reader-friendly headings and round only the displayed percentages to one decimal place. [18]

(b) Pick one of the five airlines. Filter its underlying flight records and recompute its numerator and denominator independently of the summary table. Show the arithmetic for that airline's percentage and check that it agrees with the table. [6]

(c) Suppose the editor divides the total number of known late arrivals by *all scheduled flights*. Calculate this alternative for the all-airlines row. Report how much it changes the headline rate in percentage points and explain what the alternative denominator counts. [6]

4. Pitch and illustrate your own finding [40 marks]

Choose a reader and a question you can answer from these records. Go beyond redrawing the five-airline table. For example, how does the chance of a late arrival change from morning to evening at each airport? How often does a flight that leaves at least 15 minutes late still arrive by its scheduled time? You may pursue a different question.

(a) **The pitch.** Name your reader and the comparison they care about. Define the measure, the flights you include and any cutoffs you choose. Give the number of flights behind the figure. [8]

(b) **The figure.** Make one figure in plotnine. Choose the geometry, variables, grouping and scale that bring out your comparison. Give it a headline supported by the data, label the units and make the groups easy to distinguish at the size a reader will see. [14]

(c) **The edit.** Keep a first version in the notebook. Change one design decision that affects how a reader sees the evidence. Show the final version and explain in two sentences what is easier to see and what detail, if any, is lost. A simple figure can earn full marks. [8]

(d) **The caption.** In 60–100 words, answer your question with a numerical comparison calculated in the notebook. Name the source, period and one specific limit on what the comparison can establish. [10]

What to submit

Open a pull request to `main` from your own branch. In `submissions/exercise-02/YOUR-USERNAME/`, include `exercise_02.ipynb` with the table, figure drafts and caption, the final `figure.png`, and a short `README.md` with the download and run instructions. Keep the two raw downloads on your computer. Restart the kernel and run all cells in order to check that the notebook recreates the table and saves the final figure.